Week 12
الأسبوع 12: من - إلى

Subject/:الموضوع Chemistry **Unit Title : عنوان الوحدةRevision: Structure 1–3, Reactivity 1–2** **Grade 12 DP (Year 2)**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 **Revision: Structure 1–3** **مراجعة: البنية 1–3**	• Students will apply models of particles, bonding and the classification of matter to answer exam-style questions on Structure 1, 2 and 3.	3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	• Structure overview and exam tips (RevisionDojo) • **Exam-style questions (class Microsoft Teams channel)**
Lesson 2 **الحصة الثانية** **Revision: Reactivity 1** **مراجعة: التفاعلية 1**	• Students will apply measuring enthalpy changes, energy cycles and energy from fuels to answer exam-style questions on Reactivity 1.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	• Notes: R1.1 Measuring enthalpy changes (ibchem.com) • Notes: R1.2 Energy cycles in reactions (ibchem.com) • Cheatsheets: R1.3 Energy from fuels (RevisionDojo) • **Exam-style questions (class Microsoft Teams channel)**
Lesson 3 **الحصة الثالثة** **Revision: Reactivity 2** **مراجعة: التفاعلية 2**	• Students will apply ideas about amount, rate and extent of chemical change to answer exam-style questions on Reactivity 2.	3.4 Support transfer using checklists, organizers, and recurring practice. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	• R2.3 Equilibrium topic page (RevisionDojo) • **Exam-style questions (class Microsoft Teams channel)**
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Revision resources and exam-style questions are shared on the class Microsoft Teams channel. Remind students to bring a calculator, the IB data booklet and their revision notes. [Add test or mock exam date and homework due date]		

Week # 12 الأسبوع#

Technology Integration دمج التكنولوجيا	Class Microsoft Teams channel for revision resources and exam-style questions; online notes and cheatsheets (ibchem.com, RevisionDojo).
Cross-curricular links الروابط بين المواد الدراسية	Physics: atomic structure, gases and energy. Biology: functional groups and enzyme rates. Mathematics: calculations, graphs and significant figures.
External Assessment Integration دمج التقييمات الخارجية	IB Paper 1A, 1B and 2 style questions with the data booklet and command terms. Internal Assessment links: calorimetry, rate experiments and titration.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL: Self-management (revision timetable, spaced practice), Thinking (linking ideas), Communication (command terms).
Standards المعايير	• Structure 1 – Particulate nature of matter: S1.1 introduction, S1.2 nuclear atom, S1.3 electron configurations, S1.4 the mole, S1.5 ideal gases. • Structure 2 – Bonding and structure: S2.1 ionic, S2.2 covalent, S2.3 metallic, S2.4 models to materials. Structure 3 – Classification: S3.1 periodic table, S3.2 functional groups. • Reactivity 1 – What drives chemical reactions?: R1.1 measuring enthalpy change, R1.2 energy cycles in reactions, R1.3 energy from fuels (R1.4 is HL only). • Reactivity 2 – How much, how fast and how far?: R2.1 the amount of chemical change, R2.2 the rate of chemical change, R2.3 the extent of chemical change.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Tools 1 and 3: How can experimental data from calorimetry, rate and titration investigations be used to test chemical models? Structure 2: How do bonding models help us explain the properties of materials? TOK: To what extent are models, such as the particle model and the bonding models, useful even though they are simplifications?
Success Criteria معايير النجاح **(Measurable)**	**Must (DOK 1–2):** I can recall key definitions, equations and data booklet values and use them in one-step calculations and short answers. **Should (DOK 2–3):** I can link ideas across topics, such as bonding, energetics and rate, to answer multistep exam-style questions with correct working, units and signs. **Could (DOK 3–4):** I can evaluate the limitations of models and experimental methods and justify my answers using command terms such as explain, justify and evaluate.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding التفاهم العالمي Choose an item. المجال Area: Innovation and Infrastructure التطبيق Implementation: Students apply chemical models and calculations to real-world contexts such as fuels and materials, connecting chemistry to innovation and infrastructure.